Given: Due:	Further Maths (Core Pure) HW 22	MARK SCHEME

Chapter 8: Proof by Induction

Q1. Prove by induction that, for $n \in \mathbb{Z}^+$,

$$f(n) = 8^n - 2^n$$

is divisible by 6

(6)

Students typically showed an understanding of the principal of mathematical induction and were able to obtain the first two marks in this question. Almost all students showed the function was divisible by 6 for k = 1. The majority of students then considered f $(k + 1) - m$f (k) . Those who used m = 1 or m = 2 were the most successful. In a significant number of cases the students incorrectly manipulated the algebra to write their expression as a multiple of 6. Many got stuck after forming f $(k + 1) -$ f (k)*, only gaining the first two marks. A significant minority of students lost accuracy marks by not explicitly expressing f* $(k + 1)$ explicitly as a multiple of 6. The majority of those who correctly manipulated the algebra, in general, followed through to produce a complete solution.

Question Number	Scheme	Notes	Marks
	$f(n) = 8^n - 2^n$ is divisible by 6.		
	$f(1) = 8^1 - 2^1 = 6,$	Shows that $f(1) = 6$	B1
	Assume that for $n = k$, $f(k) = 8^k - 2^k$ is divisible by 6.		
	$f(k+1) - f(k) = 8^{k+1} - 2^{k+1} - \left(8^k - 2^k\right)$	Attempt $f(k+1) - f(k)$	M1
	$= 8^k(8-1) + 2^k(1-2) = 7 \times 8^k - 2^k$		
	$= 6 \times 8^k + 8^k - 2^k \left(= 6 \times 8^k + f(k)\right)$	M1: Attempt $f(k+1) - f(k)$ as a multiple of 6 A1: rhs a correct multiple of 6	M1A1
	$f(k+1) = 6 \times 8^k + 2f(k)$	Completes to $f(k+1) =$ a multiple of 6	A1
	If the result is **true** for $n = k$, then it is now **true** for $n = k+1$. As the result has been shown to be **true** for $n = 1$, then the result is **true** for **all** n $(\in \mathbb{Z}^+.)$		A1cso
		Do not award final A if n defined incorrctly e.g. 'n is an integer' award A0	
			(6)
			Total 6
Way 2	$f(1) = 8^1 - 2^1 = 6,$	Shows that $f(1) = 6$	B1
	Assume that for $n = k$, $f(k) = 8^k - 2^k$ is divisible by 6.		
	$f(k+1) = 8^{k+1} - 2^{k+1} = 8\left(8^k - 2^k + 2^k\right) - 2.2^k$	Attempts $f(k+1)$ in terms of 2^k and 8^k	M1
	$f(k+1) = 8^{k+1} - 2^{k+1} = 8\left(f(k) + 2^k\right) - 2.2^k$	M1:Attempts $f(k+1)$ in terms of $f(k)$ A1: rhs correct and a multiple of 6	M1A1
	$f(k+1) = 8f(k) + 6.2^k$	Completes to $f(k+1) =$ a multiple of 6	A1
	If the result is **true** for $n = k$, then it is now **true** for $n = k+1$. As the result has been shown to be **true** for $n = 1$, then the result is **true** for **all** n $(\in \mathbb{Z}^+.)$		A1cso
Way 3	$f(1) = 8^1 - 2^1 = 6,$	Shows that $f(1) = 6$	B1
	Assume that for $n = k$, $f(k) = 8^k - 2^k$ is divisible by 6.		
	$f(k+1) - 8f(k) = 8^{k+1} - 2^{k+1} - 8.8^k + 8.2^k$	Attempt $f(k+1) - 8f(k)$	M1
		Any multiple m replacing 8 award M1	
	$f(k+1) - 8f(k) = 8^{k+1} - 8^{k+1} + 8.2^k - 2.2^k = 6.2^k$	M1: Attempt $f(k+1) - f(k)$ as a multiple of 6 A1: rhs a correct multiple of 6	M1A1
	$f(k+1) = 8f(k) + 6.2^k$	Completes to $f(k+1) =$ a multiple of 6	A1
		General Form for multiple m $f(k+1) = 6.8^k + (2-m)(8^k - 2^k)$	
	If the result is **true** for $n = k$, then it is now **true** for $n = k+1$. As the result has been shown to be **true** for $n = 1$, then the result is **true** for **all** n $(\in \mathbb{Z}^+.)$		A1cso

Q2. (a) Prove by induction, that for $n \in \mathbb{Z}^+$,

$$\sum_{r=1}^{n} r(2r-1) = \frac{1}{6}n(n+1)(4n-1)$$

(6)

(b) Hence, show that

$$\sum_{r=n+1}^{3n} r(2r-1) = \frac{1}{3}n(an^2+bn+c)$$

where *a*, *b* and *c* are integers to be found.

(4)

A large number of candidates knew induction well and picked up most of the marks in this question, though there was a reluctance to take out the factor (k+1) early. A common error was to show it was true for n=1, but then just substitute k+1 into the formula instead of adding the (k+1)th term to S_k. A significant number of candidates failed to identify the (k+1) in each bracket following them finding the correct factors. A minority scored no marks by trying to prove by use of the standard summation formulae from the formula book. In part (b) those who used their answer to part (a) generally did so correctly, but a minority did not correctly follow their answer to part (a). An occasional error was to write $3S_n - S_n$ instead of $S_{3n} - S_n$.

Question Number	Scheme		Marks
(a)	$\sum_{r=1}^{n} r(2r-1) = \frac{1}{6}n(n+1)(4n-1)$		
	$n=1$; LHS $= \sum_{r=1}^{1} r(2r-1) = 1$ RHS $= \frac{1}{6}(1)(2)(3) = 1$ As LHS = RHS, the summation formula is true for $n=1$.	$\frac{1}{6}(1)(2)(3)=1$ seen	B1
	Assume that the summation formula is true for $n = k$.		
	ie. $\sum_{r=1}^{k} r(2r-1) = \frac{1}{6}k(k+1)(4k-1)$.		
	With $n = k+1$ terms the summation formula becomes:		
	$\sum_{r=1}^{k+1} r(2r-1) = \underline{\frac{1}{6}k(k+1)(4k-1) + (k+1)(2(k+1)-1)}$	$S_{k+1} = S_k + u_{k+1}$ with $S_k = \frac{1}{6}k(k+1)(4k-1)$.	M1
	$= \frac{1}{6}k(k+1)(4k-1) + (k+1)(2k+1)$		
	$= \frac{1}{6}(k+1)\left(k(4k-1) + 6(2k+1)\right)$	Factorise by $\frac{1}{6}(k+1)$	dM1
	$= \frac{1}{6}(k+1)\left(4k^2+11k+6\right)$	$\left(4k^2+11k+6\right)$ or equivalent quadratic seen	A1
	$= \frac{1}{6}(k+1)(k+2)(4k+3)$		
	$= \frac{1}{6}(k+1)(k+1+1)(4(k+1)-1)$	Correct completion to S_{k+1} in terms of $k+1$ dependent on both Ms.	dM1
	If the summation formula is <u>true for</u> $n = k$, then it is shown to be <u>true for</u> $n = k+1$. As the result is <u>true for</u> $n = 1$, it is now also <u>true for all</u> n and $n \in \mathbb{Z}^+$ by mathematical induction.	Conclusion with all 4 underlined elements that can be seen anywhere in the solution	A1 cso
			[6]

Question Number	Scheme		Marks
(b)	$\sum_{r=n+1}^{3n} r(2r-1) = S_{3n} - S_n$		
	$= \frac{1}{6}.3n(3n+1)(12n-1) - \frac{1}{6}n(n+1)(4n-1)$	Use of $S_{3n} - S_n$ or $S_{3n} - S_{n+1}$ with the result from (a) used at least once. Correct un-simplified expression.	M1 A1
	$= \frac{1}{6}n\{3(3n+1)(12n-1) - (n+1)(4n-1)\}$		
	$= \frac{1}{6}n\{3(36n^2+9n-1) - (4n^2+3n-1)\}$	Factorises out $\frac{1}{6}n$ or $\frac{1}{3}n$ and an attempt to open up the brackets.	dM1
	$= \frac{1}{6}n\{108n^2+27n-3-4n^2-3n+1\}$		
	$= \frac{1}{6}n\{104n^2+24n-2\}$		
	$= \frac{1}{3}n(52n^2+12n-1)$ $\{a=52, b=12, c=-1\}$	$= \frac{1}{3}n(52n^2+12n-1)$	A1 [4] 10

Q3. (i) Prove by induction that for $n \in \mathbb{Z}^+$

$$\begin{pmatrix} 5 & -8 \\ 2 & -3 \end{pmatrix}^n = \begin{pmatrix} 4n+1 & -8n \\ 2n & 1-4n \end{pmatrix}$$

(6)

(ii) Prove by induction that for $n \in \mathbb{Z}^+$

$$f(n) = 4^{n+1} + 5^{2n-1}$$

is divisible by 21

(6)

Mathematical induction is a well-known topic and though the new specification may introduce some different induction proofs, the two in this question should have been familiar to students. Nevertheless, mathematical induction continues to be a topic that confounds many students. Of the two parts, part (i) proved more accessible.

Part (i) was a standard proof using 2 × 2 matrix multiplication and most students knew the steps well. Some did not show the $n = 1$ case clearly and lost the first mark. Some made slips in the multiplication of terms and many did not write the final matrix in terms of (k + 1), leaving the examiner to interpret that the final matrix was of the required form. The final statement was usually clear but some did not make the 'if true for $n = 1$ and $n = k$ then... clearly enough.

Other errors in part (a) included basic algebra errors and in a small number of cases attempting to find $M^{(k+1)}$ by incorrectly using Mk + M.

Part (ii) was less well answered. Most students achieved the first three marks for showing $n = 1$ is true, stating the assumption and attempting f(k + 1), but a large number struggled to go any further. Of those that did manage to correctly obtain an expression that contained f(k), the majority went on to achieve full marks, though some that used way 1 or 3 did not find a correct expression for f(k + 1) alone. All four methods were used with none standing out as being the most successful.

Question	Scheme	Marks	AOs
(i)	$n=1, \begin{pmatrix}5 & -8\\ 2 & -3\end{pmatrix}^1 = \begin{pmatrix}5 & -8\\ 2 & -3\end{pmatrix}, \begin{pmatrix}4\times1+1 & -8(1)\\ 2\times1 & 1-4(1)\end{pmatrix} = \begin{pmatrix}5 & -8\\ 2 & -3\end{pmatrix}$ So the result is true for $n = 1$	B1	2.2a
	Assume true for $n=k$ so $\begin{pmatrix}5 & -8\\ 2 & -3\end{pmatrix}^k = \begin{pmatrix}4k+1 & -8k\\ 2k & 1-4k\end{pmatrix}$	M1	2.4
	$\begin{pmatrix}5 & -8\\ 2 & -3\end{pmatrix}^{k+1} = \begin{pmatrix}4k+1 & -8k\\ 2k & 1-4k\end{pmatrix}\begin{pmatrix}5 & -8\\ 2 & -3\end{pmatrix}$ or $\begin{pmatrix}5 & -8\\ 2 & -3\end{pmatrix}^{k+1} = \begin{pmatrix}5 & -8\\ 2 & -3\end{pmatrix}\begin{pmatrix}4k+1 & -8k\\ 2k & 1-4k\end{pmatrix}$	M1	1.1b
	$\begin{pmatrix}4k+1 & -8k\\ 2k & 1-4k\end{pmatrix}\begin{pmatrix}5 & -8\\ 2 & -3\end{pmatrix} = \begin{pmatrix}5(4k+1)-16k & -8(4k+1)+24k\\ 10k+2(1-4k) & -16k-3(1-4k)\end{pmatrix}$ or $\begin{pmatrix}5 & -8\\ 2 & -3\end{pmatrix}\begin{pmatrix}4k+1 & -8k\\ 2k & 1-4k\end{pmatrix} = \begin{pmatrix}5(4k+1)-16k & -40k-8(1-4k)\\ 2(1+4k)-6k & -16k-3(1-4k)\end{pmatrix}$	A1	1.1b
	$= \begin{pmatrix}4(k+1)+1 & -8(k+1)\\ 2(k+1) & 1-4(k+1)\end{pmatrix}$	A1	2.1
	<u>If true for $n=k$</u> then <u>true for $n=k+1$</u>, <u>true for $n=1$</u> so <u>true for all (positive integers) n</u> (Allow "for all values")	A1	2.4
		(6)	

(ii) Way 1	$f(k+1)-f(k)$		
	When $n=1$, $4^{n+1}+5^{2n-1}=16+5=21$ so the statement is true for $n = 1$	B1	2.2a
	Assume true for $n=k$ so $4^{k+1}+5^{2k-1}$ is divisible by 21	M1	2.4
	$f(k+1)-f(k)=4^{k+2}+5^{2k+1}-4^{k+1}-5^{2k-1}$	M1	2.1
	$=4\times4^{k+1}+25\times5^{2k-1}-4^{k+1}-5^{2k-1}$		
	$=3f(k)+21\times5^{2k-1}$ or e.g. $=24f(k)-21\times4^{k+1}$	A1	1.1b
	$f(k+1)=4f(k)+21\times5^{2k-1}$ or e.g. $f(k+1)=25f(k)-21\times4^{k+1}$	A1	1.1b
	<u>If true for $n=k$</u> then <u>true for $n=k+1$</u>, <u>true for $n=1$</u> so <u>true for all (positive integers) n</u> (Allow "for all values")	A1	2.4
		(6)	

(ii) Way 2	f(k + 1)		
	When $n = 1,\ 4^{n+1} + 5^{2n-1} = 16 + 5 = 21$ so the statement is true for $n = 1$	B1	2.2a
	Assume true for $n = k$ so $4^{k+1} + 5^{2k-1}$ is divisible by 21	M1	2.4
	$\mathrm{f}(k+1) = 4^{k+1+1} + 5^{2(k+1)-1}$	M1	2.1
	$\mathrm{f}(k+1) = 4 \times 4^{k+1} + 5^{2k+1} = 4 \times 4^{k+1} + 4 \times 5^{2k-1} + 25 \times 5^{2k-1} - 4 \times 5^{2k-1}$ $\mathrm{f}(k+1) = 4\mathrm{f}(k) + 21 \times 5^{2k-1}$	A1 A1	1.1b 1.1b
	<u>If true for $n = k$</u> then <u>true for $n = k + 1$</u>, <u>true for $n = 1$</u> so <u>true for all (positive integers) n</u> (Allow "for all values")	A1	2.4
		(6)	
(ii) Way 3	f(k + 1) − mf(k)		
	When $n = 1,\ 4^{n+1} + 5^{2n-1} = 16 + 5 = 21$ so the statement is true for $n = 1$	B1	2.2a
	Assume true for $n = k$ so $4^{k+1} + 5^{2k-1}$ is divisible by 21	M1	2.4
	$\mathrm{f}(k+1) - m\mathrm{f}(k) = 4^{k+2} + 5^{2k+1} - m\left(4^{k+1} + 5^{2k-1}\right)$	M1	2.1
	$= (4-m)4^{k+1} + 5^{2k+1} - m \times 5^{2k-1}$ $= (4-m)\left(4^{k+1} + 5^{2k-1}\right) + 21 \times 5^{2k-1}$	A1	1.1b
	$= (4-m)\left(4^{k+1} + 5^{2k-1}\right) + 21 \times 5^{2k-1} + m\mathrm{f}(k)$	A1	1.1b
	<u>If true for $n = k$</u> then <u>true for $n = k + 1$</u>, <u>true for $n = 1$</u> so <u>true for all (positive integers) n</u> (Allow "for all values")	A1	2.4
		(6)	
(ii) Way 4	f(k) = 21M		
	When $n = 1,\ 4^{n+1} + 5^{2n-1} = 16 + 5 = 21$ so the statement is true for $n = 1$	B1	2.2a
	Assume true for $n = k$ so $4^{k+1} + 5^{2k-1} = 21M$	M1	2.4
	$\mathrm{f}(k+1) = 4^{k+1+1} + 5^{2(k+1)-1}$	M1	2.1
	$\mathrm{f}(k+1) = 4 \times 4^{k+1} + 5^{2k+1} = 4\left(21M - 5^{2k-1}\right) + 5^{2k+1}$ $\mathrm{f}(k+1) = 84M + 21 \times 5^{2k-1}$	A1 A1	1.1b 1.1b
	<u>If true for $n = k$</u> then <u>true for $n = k + 1$</u>, <u>true for $n = 1$</u> so <u>true for all (positive integers) n</u> (Allow "for all values")	A1	2.4
		(6)	
			(12 marks)

Notes

(i)

B1: Shows that the result holds for $n = 1$. Must see **substitution** into the rhs.

The minimum would be: $\begin{pmatrix} 4+1 & -8 \\ 2 & 1-4 \end{pmatrix} = \begin{pmatrix} 5 & -8 \\ 2 & -3 \end{pmatrix}$.

M1: Makes a statement that assumes the result is true for some value of n (Assume (true for) $n = k$ is sufficient – note that this may be recovered in their conclusion if they say e.g. if true for $n = k$ then … etc.)

M1: Sets up a correct multiplication statement either way round

A1: Achieves a correct un-simplified matrix

A1: Reaches a correct simplified matrix with no errors **<u>and the correct un-simplified matrix seen previously.</u>** Note that the simplified result may be proved by equivalence.

A1: Correct conclusion. This mark is dependent on all previous marks apart from the B mark. It is gained by conveying the ideas of **all** four underlined points **either** at the end of their solution **or** as a narrative in their solution.

(ii) **Way 1**

B1: Shows that f(1) = 21

M1: Makes a statement that assumes the result is true for some value of n (Assume (true for) $n = k$ is sufficient – note that this may be recovered in their conclusion if they say e.g. if true for $n = k$ then … etc.)

M1: Attempts f(k + 1) – f(k) or equivalent work

A1: Achieves a correct expression for f(k + 1) – f(k) in terms of f(k)

A1: Reaches a correct expression for f(k + 1) in terms of f(k)

A1: Correct conclusion. This mark is dependent on all previous marks apart from the B mark. It is gained by conveying the ideas of **all** four underlined points **either** at the end of their solution **or** as a narrative in their solution.

Way 2
B1: Shows that $f(1) = 21$
M1: Makes a statement that assumes the result is true for some value of n (Assume (true for) $n = k$ is sufficient – note that this may be recovered in their conclusion if they say e.g. if true for $n = k$ then … etc.)
M1: Attempts $f(k+1)$
A1: Correctly obtains $4f(k)$ **or** $21\times 5^{2k-1}$
A1: Reaches a correct expression for $f(k+1)$ in terms of $f(k)$
A1: Correct conclusion. This mark is dependent on all previous marks apart from the B mark. It is gained by conveying the ideas of **all** four underlined points **either** at the end of their solution **or** as a narrative in their solution.

Way 3
B1: Shows that $f(1) = 21$
M1: Makes a statement that assumes the result is true for some value of n (Assume (true for) $n = k$ is sufficient – note that this may be recovered in their conclusion if they say e.g. if true for $n = k$ then … etc.)
M1: Attempts $f(k+1) - mf(k)$
A1: Achieves a correct expression for $f(k+1) - mf(k)$ in terms of $f(k)$
A1: Reaches a correct expression for $f(k+1)$ in terms of $f(k)$
A1: Correct conclusion. This mark is dependent on all previous marks apart from the B mark. It is gained by conveying the ideas of **all** four underlined points **either** at the end of their solution **or** as a narrative in their solution.

Way 4
B1: Shows that $f(1) = 21$
M1: Makes a statement that assumes the result is true for some value of n (Assume (true for) $n = k$ is sufficient – note that this may be recovered in their conclusion if they say e.g. if true for $n = k$ then … etc.)
M1: Attempts $f(k+1)$
A1: Correctly obtains $84M$ **or** $21\times 5^{2k-1}$
A1: Reaches a correct expression for $f(k+1)$ in terms of M and 5^{2k-1}
A1: Correct conclusion. This mark is dependent on all previous marks apart from the B mark. It is gained by conveying the ideas of **all** four underlined points **either** at the end of their solution **or** as a narrative in their solution.

Q4. (i) Prove by induction that, for $n \in \mathbb{Z}^+$.

$$\begin{pmatrix} 1 & 0 \\ -1 & 5 \end{pmatrix}^n = \begin{pmatrix} 1 & 0 \\ -\frac{1}{4}\left(5^n - 1\right) & 5^n \end{pmatrix}$$

(6)

(ii) Prove by induction that, for $n \in \mathbb{Z}^+$.

$$\sum_{r=1}^{n} (2r-1)^2 = \frac{1}{3}n\left(4n^2 - 1\right)$$

(6)

Most students were able to demonstrate their understanding of the concept of proof by induction. A large number of students produced clear, well set out solutions, with a correct conclusion.
Almost all students started by attempting to check for n = 1, but often neglected to state that they had shown it to be true next to their working at this stage. Most students correctly assumed the statement for $n = k$ and knew they had to do something for $n = k + 1$, but not all were sure how to proceed.
Matrix multiplication in part (i) was generally well executed, although many students struggled to rearrange their answer to resemble what they were trying to prove. Some missed out an intermediate stage altogether. Students who took $(2k + 1)$ out as a factor at the beginning were generally successful. More students, however, multiplied everything out and then took $(k + 1)$ out as a factor. Some students arrived at the expression with 3 linear factors and then struggled to complete the proof. Very few students successfully started at both ends arriving at the correct cubic in the middle.

Question Number	Scheme	Marks
(i)	If $n=1$, $\begin{pmatrix}1 & 0\\ -1 & 5\end{pmatrix}^1 = \begin{pmatrix}1 & 0\\ -\frac{1}{4}(5^1-1) & 5^1\end{pmatrix}$ so **true** for $n = 1$	B1
	Assume result true for $n=k$	
	$\begin{pmatrix}1 & 0\\ -1 & 5\end{pmatrix}^{k+1} = \begin{pmatrix}1 & 0\\ -\frac{1}{4}(5^k-1) & 5^k\end{pmatrix}\begin{pmatrix}1 & 0\\ -1 & 5\end{pmatrix}$ or $\begin{pmatrix}1 & 0\\ -1 & 5\end{pmatrix}\begin{pmatrix}1 & 0\\ -\frac{1}{4}(5^k-1) & 5^k\end{pmatrix}$	M1
	$\begin{pmatrix}1 & 0\\ -1 & 5\end{pmatrix}^{k+1} = \begin{pmatrix}1 & 0\\ -\frac{1}{4}(5^k-1)-5^k & 5\times5^k\end{pmatrix}$ or $\begin{pmatrix}1 & 0\\ -1-5.\frac{1}{4}(5^k-1) & 5\times5^k\end{pmatrix}$	M1 A1
	$=\begin{pmatrix}1 & 0\\ -\frac{1}{4}5^k+\frac{1}{4}-5^k & 5^{k+1}\end{pmatrix}$ or $\begin{pmatrix}1 & 0\\ -1-\frac{1}{4}5^{k+1}+\frac{5}{4} & 5^{k+1}\end{pmatrix} = \begin{pmatrix}1 & 0\\ -\frac{1}{4}(5^{k+1}-1) & 5^{k+1}\end{pmatrix}$	A1
	True for $n = k + 1$ **if true** for $n = k$, (and **true** for $n = 1$) so **true** by induction for all $n \in \mathbb{Z}^+$.	A1cso (6)
(ii)	If $n=1$, $\sum_{r=1}^{n}(2r-1)^2 = 1$ and $\frac{1}{3}n(4n^2-1)=1$, so **true** for $n = 1$.	B1
	Assume result true for $n=k$ so $\sum_{r=1}^{k+1}(2r-1)^2 = \frac{1}{3}k(4k^2-1)+(2(k+1)-1)^2$	M1
	$=\sum_{r=1}^{k+1}(2r-1)^2 = \frac{1}{3}(2k+1)\{(2k^2-k)+(3(2k+1))\}$	M1 A1
	$=\frac{1}{3}(2k+1)\{(2k^2+5k+3)\}$ or $\frac{1}{3}(k+1)(4k^2+8k+3)$ or $\frac{1}{3}((2k+3)(2k^2+3k+1)\}$	
	$=\frac{1}{3}(k+1)(2k+1)(2k+3) \quad = \frac{1}{3}(k+1)(4(k+1)^2-1)$	dA1
	True for $n = k + 1$ **if true** for $n = k$, (and **true** for $n = 1$) so **true** by induction for all $n \in \mathbb{Z}^+$	A1cso (6)
		12 marks

Notes

(i) B1: Checks $n=1$ on both sides and states true for $n=1$ seen anywhere.
M1: Assumes true for $n = k$ and indicates intention to multiply power k by power 1 either way around.
M1: Multiplies matrices. Condone one slip. A1: Correct unsimplified matrix
A1: Intermediate step required cao
A1: cso Makes correct induction statement including at least statements in bold.
Statement **true** for $n = 1$ here could contribute to B1 mark earlier.
(ii) B1: Checks $n=1$ on both sides and states true for $n=1$ seen anywhere.
M1: Assumes true for $n=k$ and adds $(k+1)^{\text{th}}$ term to sum of k terms. Accept $4(k+1)^2-4(k+1)+1$ or $(2k+1)^2$ for $(k+1)^{\text{th}}$ term. M1: Factorises out a linear factor of the three possible - usually $2k+1$
A1: Correct expression with one linear and one quadratic factor.
dA1: Need to see $\frac{1}{3}(k+1)(4(k+1)^2-1)$ somewhere dependent upon previous A1.
Accept assumption plus $(k+1)^{\text{th}}$ term and $\frac{1}{3}(k+1)(4(k+1)^2-1)$ both leading to $\frac{1}{3}(4k^3+12k^2+11k+3)$ then award for expressions seen as above.
A1: cso Makes correct complete induction statement including at least statements in bold. Statement true for $n = 1$ here could contribute to B1 mark earlier.